

**FITNOURISH AI – INTELLIGENT NUTRITION &
WELLNESS ASSISTANT**

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Project Proposal Report

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Declaration

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I declare that this proposal is a result of our own work and has not been submitted for any degree or other programs at any other institution. I confirm, to the best of my knowledge, that it does not contain any material that is previously published or written material, except where due acknowledgement is made within the text of this report.

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As the supervisor of the above-mentioned students, I hereby confirm that they are undertaking their final year undergraduate research project under my supervision and guidance.

Sony
Supervisor's Signature

29/8/21
Date

Abstract

Sri Lanka is currently experiencing a dual challenge: the rising cost of living and the growing burden of non-communicable diseases such as diabetes, hypertension, and obesity. These factors make it increasingly difficult for individuals and families to maintain healthy dietary habits while staying within their financial limits. Although mobile applications for calorie counting, meal planning, and fitness monitoring are widely available, most of these solutions are designed for generic contexts. They rarely account for the local realities of fluctuating food prices, the cultural importance of staple ingredients, or the dietary restrictions required by people with specific health conditions. As a result, a gap persists between the nutritional advice that users receive and the affordability and practicality of their grocery purchasing decisions.

This research proposes the development of an AI-driven Smart Grocery List and Local Store Optimizer designed specifically for the Sri Lankan context. The system will take meal plan outputs and convert them into structured grocery lists that are cost-effective, nutritionally balanced, and sensitive to user preferences such as low-sodium or high-fiber diets. By integrating real-time grocery price data from local markets and supermarkets, the system will identify affordable purchasing options and suggest healthier alternatives where possible. An intuitive user interface will provide clear outputs, including itemized grocery costs, recommended stores, estimated savings, and nutritional insights tailored to individual health needs.

By combining artificial intelligence, machine learning, and localized data sources, this solution aims to make healthy eating both affordable and practical. Beyond reducing household expenditure on food, the system is expected to encourage sustainable dietary practices aligned with national nutrition guidelines. Ultimately, the proposed platform has the potential to empower Sri Lankan households to improve their nutritional outcomes while effectively managing financial constraints.

Keywords — Smart grocery list; Budget optimization; Local market integration; Personalized nutrition; Health-focused shopping.

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Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
NLP	Natural Language Processing
CNN	Convolutional Neural Network
RAG	Retrieval Augmented Generation
NER	Named Entity Recognition
BERT	Bidirectional Encoder Representations from Transformers
GPT	Generative Pre-Trained Transformer
SLM	Small Language Model
FCDB	Food Composition Database (Sri Lanka)
FBDG	Food-Based Dietary Guidelines (Sri Lanka)
HARTI	Hector Kobbekaduwa Agrarian Research and Training Institute
DCS	Department of Census and Statistics (Sri Lanka)
API	Application Programming Interface
NCD	Non-Communicable Disease
LP	Linear Programming

1 INTRODUCTION

1.1 Background and Literature Survey

In Sri Lanka, maintaining a balanced diet has become increasingly difficult due to rising food prices and the growing prevalence of non-communicable diseases (NCDs) such as diabetes, hypertension, and obesity [7]. Poor dietary habits, limited affordability, and the absence of localized nutritional support contribute significantly to this burden. While digital health platforms such as MyFitnessPal, Yazio, and Samsung Health offer features like calorie tracking, macronutrient analysis, and fitness monitoring, these applications are primarily designed for Western populations and fail to reflect Sri Lankan dietary practices and socio-economic realities [1]–[3].

Recent advancements in artificial intelligence (AI) and machine learning (ML) have opened new opportunities in dietary personalization. Papastratis et al. demonstrated a deep generative model that creates personalized meal plans aligned with nutritional standards [1]. NutriGen leverages large language models (LLMs) to generate diet plans tailored to user-specific constraints with improved caloric accuracy [2], while DietGlance integrates knowledge-based methods to support real-time dietary monitoring and personalized analysis [3]. These systems highlight the value of AI-driven personalization but remain limited to meal generation and nutrient analysis, without addressing affordability or localized market integration.

Other studies emphasize broader opportunities and challenges. Merdivenci et al. discussed scalability and data collection barriers that limit practical adoption [6], while Wang et al. reinforced the potential of AI in dietary interventions but noted that affordability is rarely considered [15]. Optimization-focused approaches such as ML-based clustering, price comparison, and demand forecasting further demonstrate the potential to merge nutrition with grocery optimization [16], [19], [21], [22].

AI is also transforming grocery retail. Wolniak and Skotnicka-Zasadzień highlight the influence of digital tools such as augmented reality and e-commerce platforms in shaping consumer

purchasing behavior [4]. Similarly, Alaka et al. reviews how AI-enabled grocery systems can support public health by encouraging healthier purchasing decisions [5]. However, global systems typically overlook localized pricing, affordability optimization, and real-time market integration, which are critical in developing economies like Sri Lanka.

Within Sri Lanka, several valuable resources exist but remain underutilized. The Sri Lanka Food Composition Database (FCDB) and Food-Based Dietary Guidelines (FBDGs) provide detailed information on local foods such as rice, dhal, coconut, and leafy greens [8], [9]. Market price information is regularly published by the Hector Kobbekaduwa Agrarian Research and Training Institute (HARTI) and the Department of Census and Statistics (DCS) [10], [11]. While these datasets are highly relevant, they are fragmented and not digitized into interactive frameworks, making them inaccessible to consumers and absent in existing AI-powered dietary platforms.

In summary, while AI systems have proven effective in meal planning, food recognition, and nutrient analysis [1], [3], [6], [12], [18], they lack integration with real-time grocery pricing, localized food culture, and affordability-focused optimization [16], [19], [20], [22], [23]. This creates a research gap in bridging nutritional personalization with the financial and cultural realities of Sri Lankan households.

Therefore, the aim of this research is to develop an AI-powered Smart Grocery List and Local Store Optimizer that combines personalized nutrition planning with affordability and real-time market integration, enabling Sri Lankan consumers to adopt healthier diets without exceeding their budgets.

1.2 Research Gap

The intersection between AI-driven nutrition planning and grocery shopping optimization remains underexplored, particularly in developing countries such as Sri Lanka. While global applications like MyFitnessPal, Yazio, and Samsung Health provide calorie tracking, macronutrient breakdown, and general meal planning, these platforms are primarily designed for Western contexts and fail to adapt to local dietary guidelines, food availability, and household budget constraints [1], [2], [3].

Gap 1: Lack of Grocery Price and Market Integration

Most nutrition-focused applications conclude at meal planning without translating recommendations into actionable grocery lists. Even fewer systems incorporate real-time grocery price data to guide cost-effective decision-making. This gap is especially critical in Sri Lanka, where household budgets are highly sensitive to food price fluctuations. Although the Hector Kobbekaduwa Agrarian Research and Training Institute (HARTI) and the Department of Census and Statistics (DCS) publish weekly market bulletins, no AI-powered system currently integrates these datasets with personalized dietary planning [10], [11].

Gap 2: Limited Focus on Affordability and Budget-Conscious Optimization

Many nutrition technologies emphasize health outcomes but pay little attention to affordability. For example, DietGlance and NutriGen apply machine learning to enhance nutritional intake, yet neither platform provides budget-sensitive alternatives [2], [3]. In Sri Lanka, where food inflation is a persistent issue, financial accessibility is as critical as nutritional adequacy. The absence of cost-sensitive optimization tools thus represents a major gap in existing research [16], [23], [24].

Gap 3: Lack of Localization for Sri Lankan Food Culture

Sri Lanka's dietary practices are unique, centered around staples such as rice, dhal, coconut, leafy *mallung*, and sambols. Although national resources such as the Sri Lanka Food Composition Database (FCDB) and the Food-Based Dietary Guidelines (FBDGs) provide detailed nutritional information, these remain undigitized and are not incorporated into AI-driven frameworks [8], [9]. As a result, international applications—developed primarily for Western diets—fail to capture the cultural and nutritional realities of Sri Lankan consumers [7].

Gap 4: Lack of Multi-Constraint Optimization (Health + Budget + Store)

Most existing systems optimize only one aspect, either nutrition or retail without addressing multiple factors together. Very few combine health restrictions (e.g., diabetic-friendly or low-sodium diets [6], [15]), cost optimization to fit household budgets [16], [19], [22], and real-time store pricing and availability [20], [21]. This gap highlights the need for a system that can balance all three dimensions, which the proposed Smart Grocery List and Local Store Optimizer seeks to achieve.

Feature / Gap	Global Apps (MyFitnessPal, Yazio, Samsung Health)	AI Retail Research (Wolniak et al., 2024; Mwangi et al., 2023)	Sri Lanka Databases (FCDB, FBDGs)	Proposed System
Personalized nutrition planning	Yes	Yes	Yes	Yes
Grocery price integration	No	No	No	Yes
Budget-conscious optimization	No	No	No	Yes
Sri Lankan food culture & localization	No	No	Yes	Yes
Real-time local store recommendations	No	No	No	Yes
Multi-constraint optimization (health+cost+store)	No	No	No	Yes

Table 1: Research gap comparison

For the proposed system, the solution will leverage AI/ML tools (Python, TensorFlow, NLP-based models) and PostgreSQL for database management. Data will be sourced from national resources (HARTI, DCS, FCDB, FBDGs), supermarket price catalogs, and user health profiles to enable budget-conscious, localized, and real-time grocery optimization.

1.3 Research Problem

In Sri Lanka, maintaining a healthy diet has become increasingly challenging due to rising food prices, limited nutritional awareness among consumers, and the high prevalence of non-communicable diseases (NCDs) such as diabetes, hypertension, and obesity. The World Health Organization (WHO) reports that the country continues to face a growing burden of diet-related diseases, with poor eating habits being a major contributing factor [7]. Although many individuals understand the importance of a balanced diet, they often struggle to translate this knowledge into affordable and practical grocery decisions.

Globally, digital health and nutrition platforms such as MyFitnessPal, Yazio, and Samsung Health have become popular for calorie tracking, meal planning, and diet management [1], [2]. However, these applications are predominantly developed for Western contexts and fail to reflect the Sri Lankan food environment, cultural dietary practices, and economic constraints. For example, Sri Lanka's Food-Based Dietary Guidelines (FBDGs) and the Sri Lanka Food Composition Database (FCDB) emphasize local staples such as rice, pulses, coconut, and leafy vegetables [8], [9]. Yet, these food items are rarely represented in international apps, leading to recommendations that are neither culturally appropriate nor financially accessible.

Research in AI-driven personalized nutrition has advanced in areas such as automated meal generation and food image recognition [1], [3], [4]. Despite these developments, no existing system connects AI-generated meal plans with real-time grocery pricing or budget optimization. This gap is particularly significant in Sri Lanka, where household food expenditure is highly vulnerable to frequent price fluctuations. Institutions such as the Hector Kobbekaduwa Agrarian Research and Training Institute (HARTI) and the Department of Census and Statistics (DCS) publish weekly market price reports, but these remain fragmented and are not integrated into consumer-facing nutrition platforms [10], [11].

As a result, many users face a disconnect between dietary recommendations and actual shopping practices. They may receive meal plans that are nutritionally adequate but unaffordable, or grocery lists that ignore factors such as store-level price variations, food availability, or specific health-related restrictions (e.g., low-sodium diets for hypertensive patients) [6], [15]. This leads to poor

adherence to dietary guidelines and added financial strain on households.

Therefore, the central research problem can be framed as follows: How can an AI-powered system be developed to generate personalized, affordable, and health-conscious grocery lists that integrate real-time local pricing, cultural dietary practices, and individual nutritional needs for Sri Lankan households?

2 OBJECTIVES

2.1 Main Objectives

The main objective of this research is to design and develop an AI-powered Smart Grocery List Optimization System that tailors weekly grocery lists based on the user's budget, nutritional needs, and local store availability. The system specifically targets the Sri Lankan context, where rising food costs and limited access to personalized nutritional guidance make healthy eating increasingly difficult. To achieve this, the system will integrate natural language processing (NLP) for analyzing user inputs, data preprocessing techniques for cleaning and standardizing grocery data, and optimization algorithms such as linear programming and constraint-based optimization to balance cost, nutrition, and product availability. By applying these methods, the system will dynamically generate shopping lists that are both cost-efficient and nutritionally balanced. Through real-time monitoring of local store prices, the solution aims to guide consumers toward healthier and more affordable purchasing decisions. Ultimately, this research seeks to deliver a practical, intelligent, and user-friendly tool that simplifies grocery shopping while promoting healthier dietary practices for Sri Lankan households.

2.2 Specific Objectives

2.2.1 Data collection and preprocessing

The first objective focuses on the collection and preprocessing of grocery-related data from local sources, including supermarket websites, retail store databases, and structured market surveys. The dataset will incorporate key attributes such as product names, categories, nutritional values, and pricing information. Given that raw data obtained from multiple sources is often inconsistent, duplicated, or incomplete, it will undergo a systematic preprocessing stage. This involves cleaning, normalizing, and structuring the data to ensure accuracy and consistency. High-quality data is essential, as the performance of the grocery list generation module directly depends on the reliability of its inputs. By producing a comprehensive and well-organized dataset, the system will be equipped to generate realistic,

accurate, and actionable grocery recommendations for end users.

2.2.2 Grocery list generation

The second objective is to design and implement a grocery list generation module capable of automatically converting meal plan outputs or user-defined preferences into well-structured shopping lists. This module will employ NLP-based similarity algorithms such as cosine similarity or Levenshtein distance to match meal plan ingredients with available store products. To handle multiple product alternatives, the system will incorporate rule-based and greedy selection methods that prioritize affordability, nutritional adequacy, and availability. For adjusting quantities based on household size or consumption patterns, the module will apply scaling algorithms to ensure realistic portioning. By combining these techniques, the system will generate optimized, practical, and user-friendly shopping lists that minimize manual effort while meeting both nutritional and practical requirements.

2.2.3 Dietary and nutritional filters

The third objective is to integrate dietary and nutritional filters that enable users to personalize their grocery lists according to specific health requirements and lifestyle choices. These filters will support diverse dietary needs such as low-sugar, gluten-free, vegetarian, or high-protein diets. By analyzing nutritional labels and product information, the system will be able to identify and recommend healthier substitutes where available. This functionality ensures that the system extends beyond simple grocery list generation, actively promoting nutritionally balanced and health-conscious decision-making. For instance, if a user indicates a preference to reduce sugar intake, the system will automatically suggest alternative items with lower sugar content, thereby aligning grocery recommendations with the user's long-term health goals.

2.2.4 Real-time budget optimization

The fourth objective is to develop a real-time budget optimization module capable of adapting grocery lists to dynamic price changes across stores and brands. Since grocery prices fluctuate regularly, affordability can be a significant barrier for many households. To mitigate this, the system will integrate real-time price comparison mechanisms and automatically adjust shopping lists to remain within the user's specified budget. This optimization may involve recommending lower-cost substitutes, adjusting product quantities, or prioritizing essential items over non-essential ones. Advanced algorithmic approaches will be employed to balance cost efficiency with nutritional adequacy, ensuring that users are able to manage expenses effectively without compromising on health outcomes.

2.2.5 User interface development

The fifth objective is to design and implement a mobile application interface that enables seamless interaction between the user and the system. This mobile app will present an optimized grocery list along with key details such as individual product prices, total estimated cost, store recommendations, and suggested alternatives. The design will emphasize clarity, accessibility, and ease of navigation, ensuring that users with varying levels of technical proficiency can effectively utilize the system. Additional features such as filtering, sorting, and exporting grocery lists will be incorporated to enhance practicality in everyday use. By delivering the interface through a mobile platform, the system ensures accessibility on-the-go, making it convenient for users to apply recommendations directly during grocery shopping. A well-structured and intuitive interface is essential for promoting adoption and providing a smooth, efficient, and engaging user experience.

2.2.6 System validation and evaluation

The final objective is to validate and evaluate the system to ensure its effectiveness and practicality in real-world contexts. The evaluation will be carried out in two stages: technical evaluation and user evaluation.

Technical Evaluation – This will focus on measuring the system’s performance and accuracy. Key aspects include:

- Accuracy of grocery list generation.
- Efficiency of the budget optimization module, including percentage cost reduction compared to baseline lists.
- Response time for generating optimized grocery lists.
- Reliability of dietary and nutritional filtering.

User Evaluation – This will focus on usability and overall user satisfaction. A group of test users will interact with the mobile application, and feedback will be collected through surveys, questionnaires, and usability testing. Key aspects include:

- Ease of use and clarity of the mobile interface.
- Practicality and relevance of the generated grocery lists in real-world shopping.
- Perceived usefulness of store comparisons and cost-saving recommendations.

By combining technical metrics with user-centered insights, the evaluation will provide a comprehensive understanding of both the functional performance and the real-world applicability of the system.

3 METHODOLOGY

This section outlines the methodology followed for the development of FitNourish AI – Intelligent Nutrition & Wellness Assistant. The system is designed as an integrated AI-powered platform that provides personalized dietary recommendations, real-time grocery optimization, meal analysis, and health monitoring. Each component in the system contributes to the overall goal of helping users achieve healthier eating habits while staying within their financial and medical constraints.

3.1 Overall System Diagram

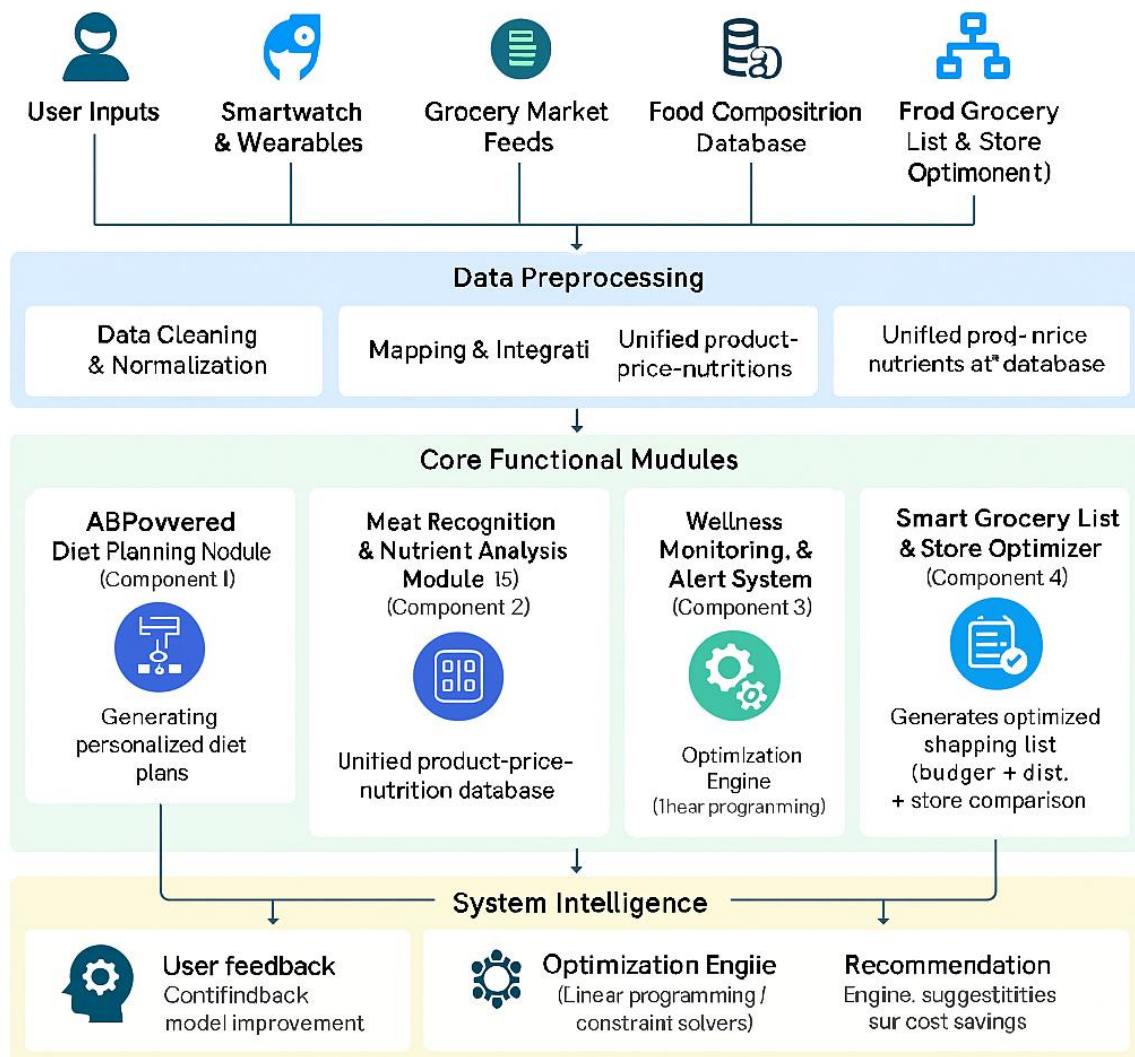


Figure 3.1: Overall System Diagram

The overall system architecture of FitNourish AI – Intelligent Nutrition & Wellness Assistant is built around four interdependent modules. Each module addresses a critical aspect of nutrition and wellness.

Data Collection and Preprocessing

The system begins with data acquisition. Inputs include user health profiles, smartwatch biometrics (e.g., heart rate, blood pressure, and sleep data), grocery market information (HARTI, DCS, and supermarket APIs), and nutritional datasets from the Sri Lanka Food Composition Database (FCDB). These data streams are aggregated, standardized, and stored for analysis.

AI-Powered Diet Planning Module

This module uses machine learning models such as Decision Trees and Random Forests to generate personalized diet plans. Recommendations are refined every two weeks. Updates are guided by user health trends and biometric indicators.

Meal Recognition and Nutrient Analysis Module

Computer vision models (e.g., Convolutional Neural Networks – CNNs) analyze food images uploaded by users. The system identifies ingredients, estimates calories and nutrient values, and recommends healthier alternatives.

Wellness Monitoring and Alert System

This module applies time-series forecasting and anomaly detection to monitor biometric and dietary data. It predicts potential health risks and generates real-time alerts when irregular patterns are detected.

Smart Grocery List and Store Optimizer

Diet plans are transformed into cost-sensitive grocery lists. The module applies dietary filters and compares real-time prices across local markets. It then suggests affordable and health-conscious alternatives.

User-Friendly Interface

All outputs are presented through a mobile-friendly interface. Users can access diet plans, receive alerts, analyze meals, generate optimized grocery lists, and track long-term progress.

By integrating these modules, FitNourish AI provides a practical solution that balances nutritional well-being with financial accessibility. The system is specifically designed to address the socio-economic and health challenges of Sri Lanka.

Sri Lanka.

3.2 Component Specific System Diagram

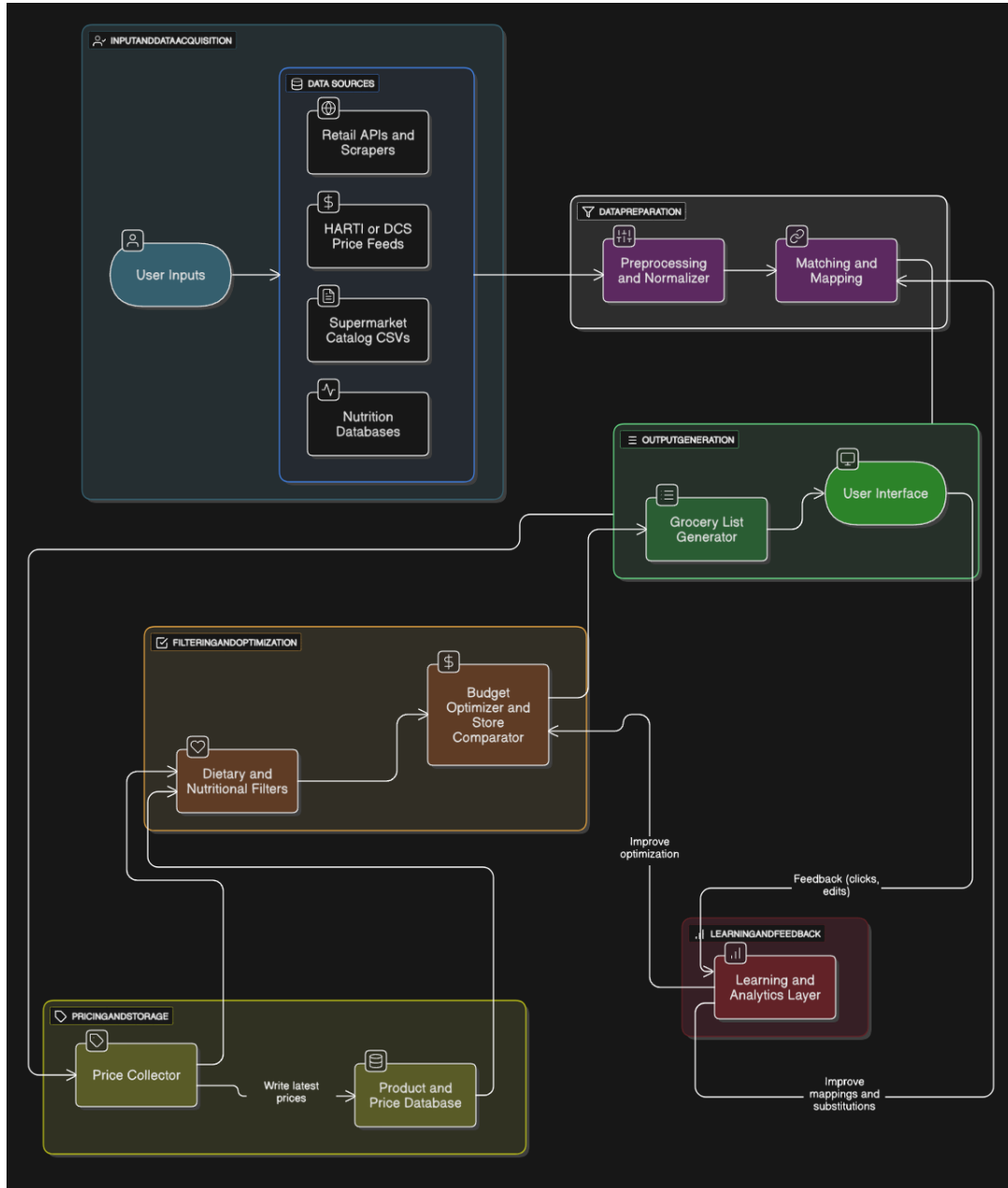


Figure 3.2: Component specific system diagram

As illustrated in the diagram above, the proposed system is structured into a series of interconnected components that collectively ensure the generation of an optimized, budget-conscious grocery list while maintaining nutritional balance. Each stage is crucial, with the proper functioning of earlier layers providing the foundation for accurate filtering, optimization, and final user outputs. The modular design also allows for clear data flow, continuous feedback, and

iterative improvements, ensuring that the system remains reliable, adaptive, and user centered.

Input and Data Acquisition:

The process begins with the integration of data from diverse sources. User-provided inputs are combined with external feeds, including supermarket APIs, web scrapers, CSV catalogues, and official datasets such as the Hector Kobbekaduwa Agrarian Research and Training Institute (HARTI) and the Department of Census and Statistics (DCS) bulletins. Nutritional data is incorporated from the Sri Lanka Food Composition Database (FCDB). Metrics Tracked: Data completeness (coverage of local products and stores), data freshness (update frequency of APIs/feeds).

Data Preparation:

The raw inputs undergo cleaning and normalization pipelines to address inconsistencies, remove duplicates, and ensure cross-format compatibility. Matching and mapping algorithms align product data with nutritional standards and user preferences, creating a structured dataset for further processing. Metrics Tracked: Data accuracy (error rate after cleaning), processing speed (time taken for preprocessing).

Pricing and Storage:

A dedicated price collection mechanism continuously updates the database with real-time retail and wholesale values. The Product and Price Repository ensures optimization decisions are based on current information. Metrics Tracked: Update latency (time gap between market price change and database update), storage efficiency (query response time)

Filtering and Optimization:

Once structured data is available, dietary and nutritional filters are applied to reflect user-specific restrictions (e.g., diabetic-friendly, low-sodium). The Budget Optimizer and Store Comparator employ algorithms to minimize overall costs while preserving nutritional adequacy. Metrics Tracked: Optimization accuracy (nutritional adequacy score), cost efficiency (percentage cost reduction vs baseline), execution speed (time to generate results).

Output Generation:

The Grocery List Generator produces an actionable shopping list, displayed through the User Interface with itemized costs, store recommendations, and compliance indicators. Features such as sorting and filtering improve usability. Metrics Tracked: Usability score (from user testing), response time (time to generate and display list), clarity (measured via user satisfaction surveys).

Learning and Feedback:

The Learning and Analytics Layer captures user interactions, purchasing outcomes, and satisfaction. Feedback loops are reintegrated to refine filtering, prediction, and optimization models over time. Metrics Tracked: User adherence rate (percentage of lists followed), prediction accuracy (alignment between recommendations and actual purchases), improvement rate (accuracy gain over iterations).

3.3 Data Collection

The effectiveness of this component depends significantly on the quality, relevance, and timeliness of collected data. Since the system optimizes grocery lists based on budget, nutrition, and local availability, trustworthy sources are essential to ensure practical and context-specific recommendations.

To support this goal, data will be collected from diverse structured and unstructured sources to build a comprehensive product–price–nutrition database. Key sources include:

1. **HARTI and DCS Price Feeds** – These official feeds, published by the Hector Kobbekaduwa Agrarian Research and Training Institute (HARTI) and the Department of Census and Statistics (DCS), provide authoritative data on wholesale and retail food prices. They serve as crucial benchmarks for tracking regional price fluctuations in essential items such as rice, dhal, coconut, vegetables, and fish.
2. **CBSL Daily Price Reports** - The Central Bank of Sri Lanka’s Daily Price Reports offer up-to-date insights into consumer item price trends across key markets. These reports help ensure real-

time responsiveness to market movements in grocery pricing across Sri Lanka.

3. **WFP Food Prices Dataset** - The World Food Programme's pricing dataset includes food price data for Sri Lanka, covering commodities like rice, beans, fish, and sugar. Updated regularly and accessible through the Humanitarian Data Exchange, this dataset provides valuable comparative data at country and regional levels.
4. **Supermarket Catalogs (CSV/Excel files)** – Structured catalogs from local supermarkets will be collected to capture comprehensive product listings, promotions, and store-specific pricing. This will enable accurate comparison of costs and availability across different retail channels.
5. **Local Nutrition Data (FCDB & FBDGs)** – The Sri Lanka Food Composition Database (FCDB) and the Food-Based Dietary Guidelines (FBDGs) provide detailed nutritional profiles for locally consumed staples (e.g., rice varieties, jackfruit, sambols, leafy greens) and will serve as the core nutritional dataset. Where local data is lacking, global sources such as the USDA FoodData Central or FAO datasets will be used only as supplementary references.

To maintain data quality and consistency, automated pipelines will be established for continuous data retrieval and updating. Preprocessing steps will normalize formats, eliminate duplicates, and resolve inconsistencies across datasets.

By consolidating real-time price feeds (HARTI, DCS, CBSL), the WFP dataset, supermarket catalogs, and nutritional profiles into a unified database, the system ensures that optimization processes are grounded in accurate, locally relevant, and current data—especially vital for addressing Sri Lankan dietary and economic realities.

3.4 Project Requirements

3.4.1 Functional requirements

Data collection and aggregation:

The system is required to continuously collect and update grocery-related data from a variety of sources, including retail APIs, supermarket catalogs, government price feeds, and nutritional databases. By integrating these heterogeneous inputs, the system can ensure comprehensive and up-to-date coverage of product prices, availability, and nutritional values, which is essential for generating accurate and reliable recommendations.

Data normalization and mapping:

To ensure consistency and reliability, all collected data will be preprocessed, standardized, and mapped into a unified format. This process will address common inconsistencies such as unit variations, duplicate entries, and naming discrepancies. By applying normalization techniques, the system will harmonize data from diverse sources, enabling accurate comparison and categorization of grocery items. This step is critical for supporting precise price analysis, product substitution, and overall optimization within the system.

Budget optimization and store comparison:

The system will incorporate optimization algorithms such as linear programming and constraint solvers to minimize the total cost of grocery purchases while ensuring nutritional adequacy and consistency with planned meals. In addition, the system will support multi-store price comparisons, enabling users to evaluate product availability and costs across different retailers. This dual approach ensures that users receive both nutritionally balanced and cost-effective recommendations, ultimately promoting affordability without compromising health outcomes.

Dietary and nutritional filtering:

The system will provide users with the ability to apply dietary filters such as vegetarian, diabetic-friendly, or high-protein preferences, alongside nutritional constraints such as daily calorie or micronutrient limits. By incorporating these customizable parameters, the system can generate personalized grocery lists that align with individual health requirements and lifestyle choices. This feature ensures that the recommendations go beyond cost and availability, actively supporting healthier and more tailored eating practices.

Intelligent categorization:

The system will employ natural language processing (NLP)-based classification models to automatically categorize grocery items into meaningful groups such as fruits, vegetables, proteins, and grains. This automated categorization enhances the organization, filtering, and retrieval of products, ensuring that users can efficiently navigate their grocery lists. By reducing manual effort and improving system intelligence, this feature supports both usability and accuracy in generating structured shopping recommendations.

Dynamic grocery list generation:

The system will generate a structured grocery list based on the outcomes of the optimization process. This list will incorporate key details such as the selected store, total estimated cost, and nutritional breakdown of the chosen items. By presenting results in a clear and organized format, the system enables users to review, adjust, and make informed decisions about their purchases. This functionality ensures that the grocery list is not only practical and actionable but also aligned with both financial constraints and nutritional goals.

3.4.2 Non-functional requirements

Data accuracy:

The system must maintain a high level of reliability and consistency in its optimization and categorization processes. To achieve this, mechanisms such as data validation, error detection, and the use of verified nutritional datasets will be incorporated. These measures ensure that both pricing information and dietary recommendations are accurate, thereby enhancing user trust and supporting informed decision-making.

Scalability:

The system must be designed to accommodate growth in both data volume and functional complexity without a decline in performance. This includes the ability to process larger product datasets, integrate additional retail stores, and manage more sophisticated dietary rules. Furthermore, architecture should support seamless integration with future data sources and external APIs, ensuring that the solution remains adaptable and sustainable as user needs and market dynamics evolve.

Usability:

The system's interface and output must be designed to be intuitive and accessible, allowing non-technical users to interact with the application effortlessly. Users should be able to input preferences and constraints, view optimized grocery lists, and compare store options with minimal effort. To further enhance usability, the interface will provide clear visualizations of cost breakdowns and nutritional summaries, ensuring that information is easy to interpret and supports informed decision making

Performance and efficiency:

The system must ensure that optimization algorithms and related

processes operate with minimal response times, allowing users to generate grocery lists and receive recommendations quickly. Efficient execution is essential to maintain user engagement and satisfaction, particularly when handling large datasets or performing real-time price comparisons. The system should therefore be optimized for both speed and resource utilization, ensuring that results are delivered promptly without compromising accuracy or reliability.

Maintainability:

The system should be developed using modular architecture, allowing individual components such as data scrapers, dietary filters, and optimization modules to be updated, extended, or replaced independently. This design ensures that future enhancements or bug fixes can be implemented without disrupting the overall functionality. A modular approach also improves long-term sustainability, reduces maintenance costs, and supports the integration of new features or technologies as the system evolves.

Security and data privacy:

The system must ensure that all user-specific data, including budget constraints, dietary preferences, and health-related filters, is stored and processed securely. Strong data protection measures such as encryption, access control, and secure communication protocols should be applied to safeguard sensitive information from unauthorized access. Furthermore, the system should adhere to ethical standards and privacy regulations, ensuring that personal data is used solely for its intended purpose and handled in a transparent and responsible manner.

3.4.3 User requirements

- Users should be able to create a profile with dietary preferences and budget constraints.
- Users should easily add, update, or remove items from the grocery list.
- Users should be able to visualize the breakdown of their grocery spending by category.
- Users should receive suggestions for cheaper or healthier alternatives.
- Users should be able to export/download their grocery list for offline use.

3.5 Tools and Technologies

This research leverages a combination of advanced tools and cutting-edge technologies to develop and deploy the Budget-Conscious Smart Cart Optimizer. The focus is on integrating artificial intelligence, data analytics, and modern web technologies to provide an intelligent and user-friendly grocery planning system.

3.5.1 Tools

Google colab (2025 release):

Colab provides a flexible cloud-based environment for experimenting with machine learning models. It supports GPU (NVIDIA Tesla T4) and TPU acceleration for tasks such as price prediction, natural language processing (NLP), and optimization. This will be used for model prototyping and training with large datasets collected from local sources.

Visual studio code (v1.92, 2025 stable build):

VS Code will serve as the primary Integrated Development Environment (IDE). It provides debugging, Git integration, and extensions for JavaScript/TypeScript, React, and Python development.

It will be used for both backend (Node.js v20, Express v4) and frontend (React.js v18) components.

PostgreSQL (v16.2)

PostgreSQL will be used as the relational database management system for storing product data, pricing, and user profiles. Its ACID compliance ensures reliability for financial and price-sensitive information. The latest version supports JSONB for handling semi-structured data alongside traditional relational schemas, allowing flexibility in integrating heterogeneous datasets.

Trello (2025 release):

Trello will be used for project management and collaboration. Kanban boards will track development phases such as data collection, model training, UI design, and testing.

TensorFlow (v2.16.1) and Keras (v3.0.5):

These frameworks will power machine learning tasks such as price prediction, classification, and dietary optimization. TensorFlow provides GPU acceleration, while Keras offers a high-level API for rapid experimentation.

3.5.2 Languages/Technologies

Python (v3.11):

Python will be the core technology for data processing, machine learning, and optimization algorithms. Libraries like Pandas and NumPy will be used for preprocessing grocery data; TensorFlow and Keras for model building; and Scikit-learn for lightweight ML experiments.

React.js (v18.2):

React will be used for building the frontend. Its component-based architecture supports modularity and scalability. It will be integrated with Tailwind CSS (v3.4) to create a responsive and modern interface.

Optimization algorithms (Linear programming / Constraint solvers):

Budget optimization requires solving a constrained optimization problem (minimizing total cost while maximizing nutritional value and keeping meal consistency). Algorithms such as Linear Programming (via libraries like PuLP v2.8) or Constraint Solvers will be used to implement this component.

Machine learning models (NLP + Classification):

Pre-trained embeddings such as BERT (base-uncased, Hugging Face Transformers v4.41) will be used for product categorization. This enables the system to intelligently classify items (fruits, proteins, grains, etc.) while applying dietary filters like low-sodium or high-protein preferences.

3.6 Implementation

The finalized outcome of the proposed solution is an interactive web-based application designed to assist users in generating cost-effective, nutritionally balanced grocery lists. This application will integrate real-time price data, dietary filters, and optimization algorithms to provide a personalized and practical shopping experience.

Frontend

The front end of the system will be developed using React.js, ensuring a responsive and user-friendly interface. React's component-based architecture enables the creation of reusable UI elements such as budget input forms, category filters, and cost comparison dashboards. This modular design allows efficient development and future scalability. React's state management and

virtual DOM will ensure smooth updates when grocery data, prices, or optimization results change, providing a seamless and interactive user experience. Visualizations such as cost breakdown charts, nutritional summaries, and store comparisons will be implemented to help users make informed decisions quickly and easily.

Backend

The back end will be implemented using Flask, chosen for its lightweight design, flexibility, and ability to quickly integrate with machine learning models and optimization algorithms. Flask will manage the RESTful APIs that connect the front end with the system's core components, such as the budget optimizer, NLP-based categorizer, and store comparator. This ensures smooth data flow between the user interface and backend services. Flask extensibility also allows easy integration with third-party libraries, such as PuLP for linear programming and Hugging Face Transformers for NLP-based categorization, making it ideal for handling the computational requirements of the system.

Database management

The system will use PostgreSQL to manage and store products, price, nutritional, and user-related data. Its relational schema ensures data integrity while supporting complex queries, such as matching grocery items with nutritional values and comparing store-level prices. PostgreSQL's indexing, constraints, and support for time-series extensions (e.g., TimescaleDB) enable efficient handling of frequently changing datasets like grocery prices and seasonal items. With features like JSONB for semi-structured data and strong scalability options, PostgreSQL provides a reliable and high-performance foundation for real-time optimization and personalized grocery list generation.

4 EVALUATION OF THE SYSTEM

The effectiveness of the Budget-Conscious Smart Cart Optimizer will be assessed through a combination of technical evaluation metrics and user-focused testing. The evaluation process aims to determine whether the system successfully delivers accurate, cost-efficient, and nutritionally balanced grocery recommendations while remaining scalable, efficient, and user-friendly.cases.

Cost optimization accuracy

A key measure of success is the system’s ability to reduce grocery costs without compromising nutritional adequacy or meal consistency. To evaluate this, optimized shopping lists will be compared against baseline lists generated without budget optimization. Metrics such as percentage cost reduction (target $\geq 15\text{--}20\%$), nutritional coverage ($\geq 90\%$ of dietary requirements met), and alignment with user-defined dietary filters ($\geq 95\%$ compliance) will be calculated. These benchmarks will ensure that the optimization engine provides both economic benefits and health-focused accuracy.

Categorization performance (ML evaluation)

Since the system employs NLP and classification models to categorize grocery items (e.g., fruits, vegetables, proteins, grains), their accuracy will be validated using standard machine learning metrics:

- Precision and Recall – Target $\geq 90\%$ for major food categories to ensure both accuracy and completeness in item classification.
- F1-Score – Target ≥ 0.90 to balance precision and recall for overall model effectiveness.
- Confusion Matrix Analysis – Misclassifications (e.g., mislabeling lentils as vegetables instead of proteins) will be tracked, with a goal of reducing error rates to $\leq 5\%$ across categories.

System efficiency and scalability

The system will be evaluated for its efficiency in generating optimized shopping lists within acceptable response times. Stress tests will simulate larger datasets (e.g., thousands of products across multiple stores) and multiple concurrent users to assess scalability.

Key metrics include:

- Average Response Time: Target 3 seconds for generating an optimized grocery list under normal load.
- System Throughput: Target ≥ 50 requests per minute without significant performance degradation.
- Concurrent Users: The system should reliably support at least 100 simultaneous users in the prototype stage, with the ability to scale further in production.
- Scalability Benchmark: The system should handle datasets containing 10,000+ products across 50+ stores while maintaining accuracy and response time targets.

User satisfaction and usability testing

Given the system's end-user focus, usability evaluation is essential. User Acceptance Testing (UAT) will be conducted with a sample group of students and households. Participants will evaluate:

- Clarity of categorized items and optimized recommendations.
- Practicality of shopping lists in real-world contexts.
- The ease of use and clarity of the web interface.

Feedback will be collected via surveys and questionnaires, and findings will be used to iteratively refine the system. High user satisfaction and positive usability scores will be considered critical indicators of success.

5 WORK BREAKDOWN STRUCTURE

We developed the below-mentioned work breakdown structure for a year. This mentions about the main deliverables such as reports or presentations and broke down tasks for smaller and manageable components. By making such a work breakdown structure, every important task is mapped out and is planned for. When it is properly mapped, each teammate can think about time and resource allocation effectively.

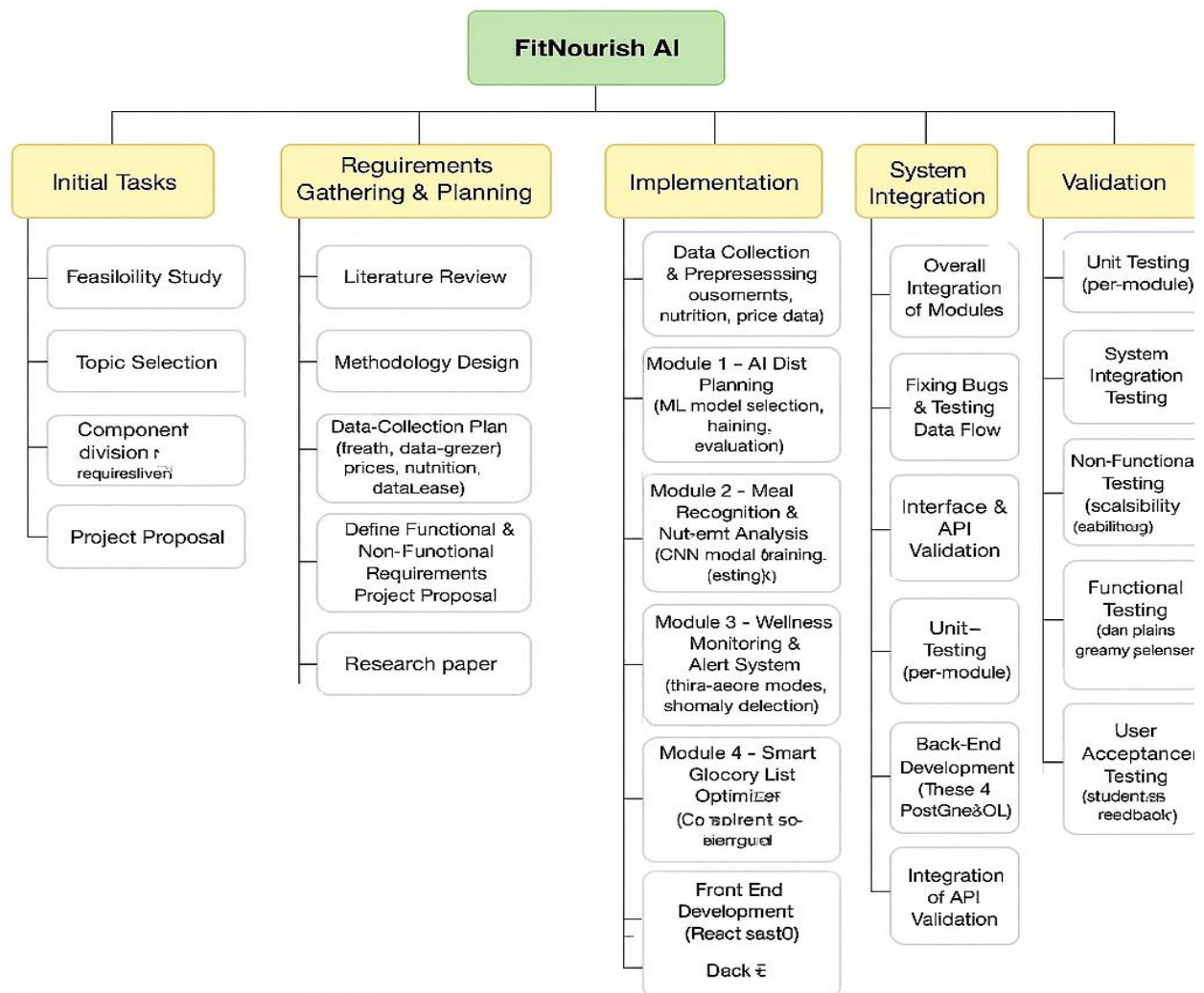


Figure 5.1: Work breakdown structure

6 GANTT CHART

A Gantt chart is a visual tool used for planning and tracking project tasks over time. It displays tasks as bars on a timeline, showing when each task starts, how long it lasts, and how tasks overlap or depend on each other. This helps you see the project's progress and meet deadlines. By Utilizing this, we could clearly identify which tasks require the most time and which require less time. This is especially useful when managing time.

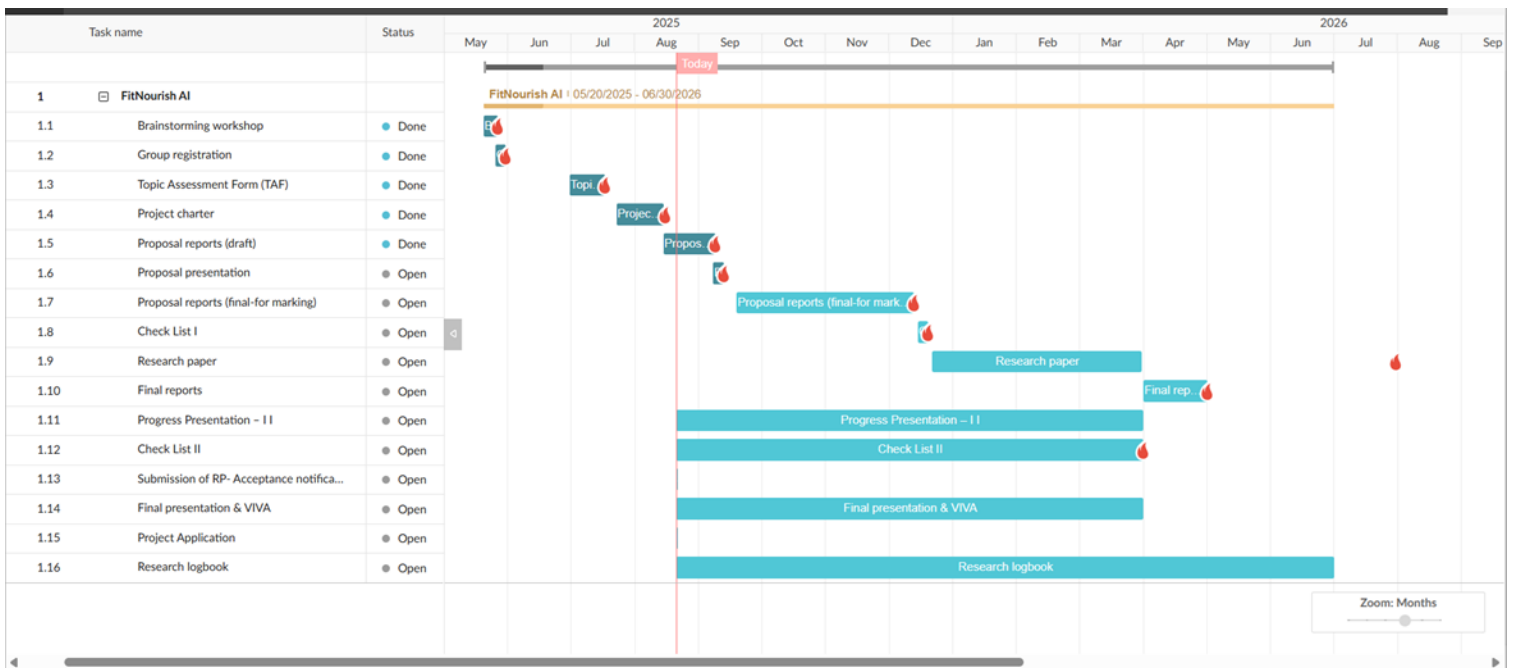


Figure 6.1: Gantt chart

7 COMMERCIALIZATION

The global health and wellness sector is experiencing rapid expansion, with the digital health market projected to reach USD 650 billion by 2025 [23]. Specifically, the AI in personalized nutrition market alone is valued at USD 3.66 billion in 2024, with expectations to grow to USD 10.37 billion by 2029 at a CAGR of 23.2% [23]. In Sri Lanka, this trend is equally relevant, as rising food prices and the growing prevalence of chronic non-communicable diseases (NCDs) such as diabetes, hypertension, and obesity [7] have created an urgent demand for affordable and accessible technology-driven solutions.

FitNourish AI directly addresses this need by providing an AI-powered nutrition and wellness assistant that uniquely integrates personalized dietary planning with real-time grocery price optimization. This dual focus on health outcomes and financial accessibility provides a strong competitive edge in the local and regional market.

Target Audience

- Primary users: Health-conscious individuals, households balancing affordability with nutrition, and patients managing diet-related conditions.
- Secondary users: Dietitians, nutritionists, and healthcare providers, who can utilize the platform for evidence-based recommendations.
- Business partners: Retailers and supermarkets, who benefit from directed consumer traffic through integrated price feeds and store recommendations.

Market Opportunity in Sri Lanka

- The fitness and wellness industry in Sri Lanka is expected to grow steadily, fueled by smartphone penetration (over 60% internet penetration in 2024) and a health-aware middle class.

- Household spending on food accounts for 34–38% of total expenditure (Department of Census and Statistics, 2024), highlighting the importance of budget-conscious dietary solutions.
- The rising popularity of online grocery platforms (e.g., Keells Online, Glomark, and PickMe Food) indicates increasing digital adoption in everyday food purchasing.

Existing Competitors

- Global platforms like *MyFitnessPal*, *Yazio*, and *Samsung Health* provide calorie tracking and diet management but lack local food databases and real-time price integration [1], [2], [3].
- Local solutions remain limited, with online grocery platforms such as *Keells Online* and *Glomark* offering convenience but no AI-driven nutritional personalization or optimization.
- Thus, FitNourish AI fills a unique gap by combining nutrition, affordability, and store-level intelligence in one integrated solution.

Potential Risks and Mitigation

- User Adoption Risk: Low digital literacy among certain rural populations may slow uptake. *Mitigation:* Begin with urban and university-based pilot projects before rural expansion.
- Competitive Risk: Established global apps may enter the local market. *Mitigation:* Strong localization (Sri Lankan foods, HARTI/DCS integration) will provide a defensible niche.
- Data Privacy and Trust: Handling user health and dietary data raises privacy concerns. *Mitigation:* Ensure compliance with international data protection practices (GDPR-inspired frameworks) and provide transparent consent options.

- **Cost of Scaling:** Cloud hosting and API integration may increase operational costs.
Mitigation: Adopt phased rollout with hybrid cloud solutions to balance cost and performance.

Commercialization Strategy

The system will launch through subscription-based models with free basic features and premium upgrades (advanced analytics, health trend reports). B2B models will target partnerships with supermarkets, wearable device providers, and healthcare institutions. Government collaborations (e.g., with the Ministry of Health or the Health Promotion Bureau) can further strengthen adoption.

8 BUDGET AND JUSTIFICATION

A preliminary budget has been prepared to cover the essential resources required for the successful implementation and evaluation of the proposed system. Table 1 summarizes the anticipated costs.

Category	Description	Estimated Cost (LKR)
Hardware / Devices	Samsung Galaxy Watch 6 – for biometric data acquisition (heartrate, SpO ₂ , sleep, stress monitoring).	50,000
Software / Tools	Python, TensorFlow, PyTorch, Google Collab (open source / free tiers).	0
Cloud / Hosting	GPU-enabled cloud resources (Google Collab Pro / AWS EC2) for model training.	6,000
Datasets	Publicly available datasets (free) and premium access where necessary.	2,000
Documentation & Printing	Report preparation, printing, and binding.	3,000
Miscellaneous	Internet, electricity, and contingency costs.	4,000
Total Estimated Budget		65,000

Table 2: Budget and justification

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Project Proposal Report

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